

LeadAI

Inbound Lead Enrichment & Automation Platform

GTM Engineer Practical Assignment : EliseAI

Section 1: Executive Summary

What LeadAI Is

LeadAI is an inbound lead enrichment and automation platform built for EliseAI's sales team. When a new lead arrives with only basic info name, email, company, and property address then LeadAI automatically researches the prospect, scores their fit across five dimensions, and generates a personalized outreach email ready to send. What currently takes an SDR 45 minutes of manual research happens in under 90 seconds.

The Problem It Solves

EliseAI SDRs face three research tasks on every inbound lead before they can make first contact:

- ICP qualification: Is this company actually a residential property manager? Many leads are real estate agents, retail property managers, or non-residential operators ,wasted effort if contacted.
- Pain research: What operational challenge is this company experiencing right now? Hiring pressure, tech gaps, expansion, market pressure , these are the signals that make outreach relevant.
- Personalized outreach: Writing a first email that references specific, verifiable facts about the prospect's situation, not a generic template.

LeadAI automates all three using public APIs and AI agents, freeing SDRs to focus exclusively on what only humans can do: building relationships and closing deals.

What Was Built

8 public APIs	Exa AI, SEC EDGAR, Adzuna Jobs, Census ACS, FRED (Federal Reserve), HUD Fair Market Rents, NewsAPI, WalkScore
7 AI agents	Identity, Company, Market, Property, Values, Scoring, Outreach , running as a LangGraph graph with parallel fan-out
1 agentic loop	Company agent uses Claude Sonnet with native tool calling. Claude plans its own research, decides which tools to call, and stops when it has enough data
Full stack	FastAPI backend, React + Vite frontend, PostgreSQL, LangGraph orchestrator, SSE live streaming, APScheduler
16 test leads	Enterprise public REITs, large private operators, student housing, affordable housing, single-family rental, and intentional disqualifications
Built in 5 days	Using free-tier APIs only — architecture designed for clear production upgrade path

Key Metrics : 16 Lead Test

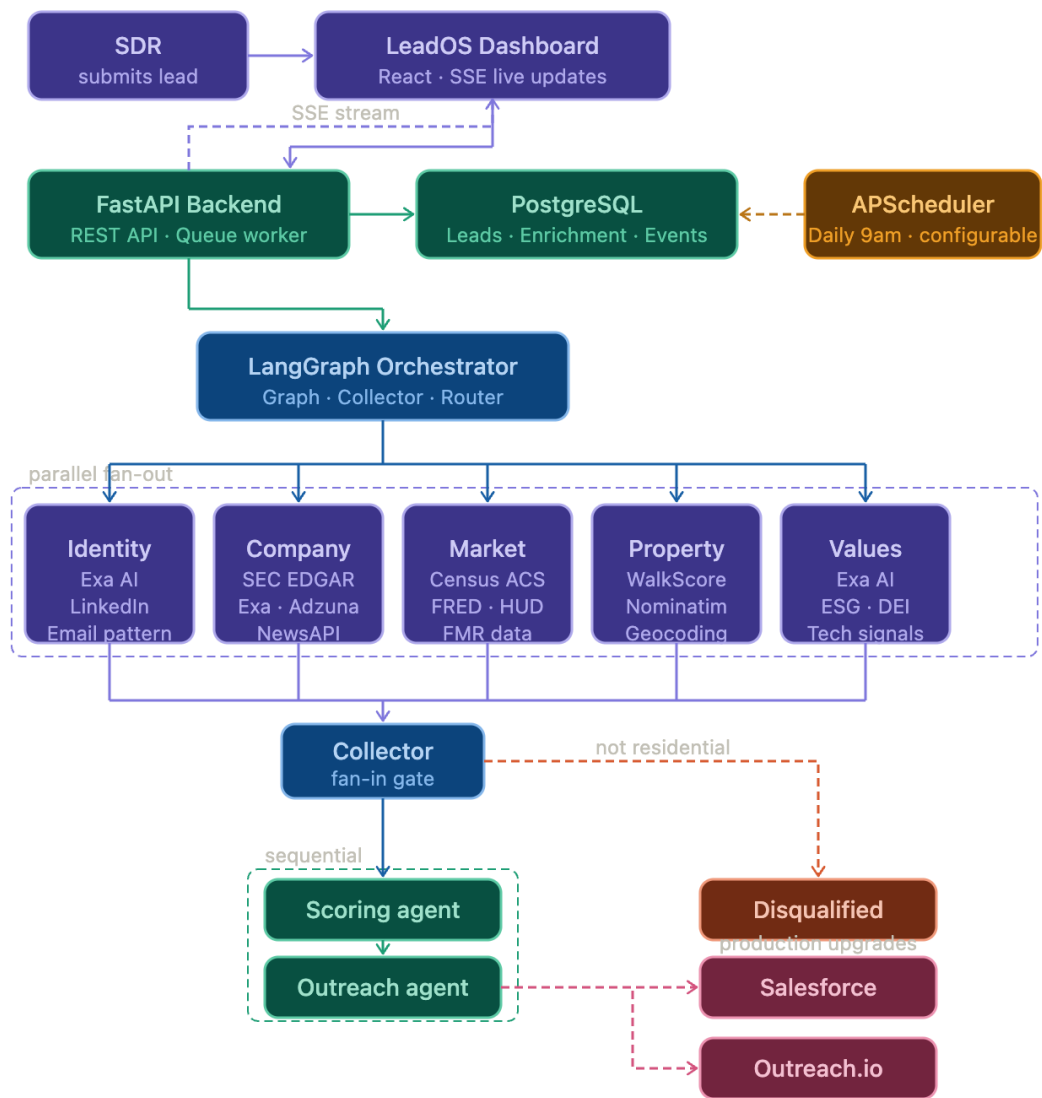
16 leads tested	10 correctly scored	6 correctly disqualified	90s avg pipeline time
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Highlight Results

Lead	Company	Score	Result
Scott Berka	Greystar	85 — HOT	3 agentic iterations, 1.1M units extracted
Trevor Samios	WinnCompanies	87 — HOT	Affordable housing correctly identified
Joe Racanelli	Lincoln Property	72 — WARM	Private enterprise, Dallas market
John Wendel	McDonald's	Disqualified	SIC 5812 — 15 seconds, no wasted cost
Nicole Collins	Equinix	Disqualified	Data center REIT — not residential
Ryan McCoy	Real McCoy Group	Disqualified	Real estate agent, not property manager

Section 2: Architecture Overview

LeadAI has four layers that work together: a React frontend, a FastAPI backend with PostgreSQL, a LangGraph pipeline orchestrator, and 8 external APIs. The diagram below shows the complete system.



■ Enrichment agents ■ Backend / sequential ■ Orchestration ■ Disqualification ■ Production

The Four Layers

Frontend	React + Vite dashboard. Lead submission form, live agent pipeline graph (React Flow), scoring methodology page, settings page for scheduler timing. SSE connection shows live agent steps as they run.
Backend	FastAPI with PostgreSQL. REST API + SSE streaming endpoint. DB-backed queue, one pipeline runs at a time, others wait. APScheduler for daily 9am trigger (configurable from Settings page). Pipeline events persisted to DB for SSE replay on browser refresh.
Orchestrator	LangGraph graph-based pipeline. 5 enrichment agents fan out in parallel from START. Collector gate waits for all 5, checks is_residential, routes to Scoring or Disqualified. Scoring -> Outreach sequential. All agent completions emit SSE events + persist to DB incrementally.
External APIs	8 public APIs , free tier for demo, clear paid upgrade path for production. Claude Sonnet for 3 LLM calls per lead: company agent agentic loop (1-4 iterations), scoring insights, outreach email generation.

The 7 Agents

Agent	Runs	Data Sources	Output
Identity	Parallel	Exa AI LinkedIn · Email pattern	Title, seniority, DM probability, LinkedIn URL
Company	Parallel	SEC EDGAR · Exa · Adzuna · NewsAPI	Size, units, revenue, tech stack, hiring signals
Market	Parallel	Census ACS · FRED · HUD FMR	Leasing intensity, product line, market health
Property	Parallel	WalkScore · OpenStreetMap Nominatim	Walk/transit/bike score, urban classification
Values	Parallel	Exa AI (3 searches)	ESG signals, AI interest, mission statement
Scoring	Sequential	All 5 agent outputs	0-100 score, tier, insights, objection prep
Outreach	Sequential	Score + enrichment data	Personalized email, subject, talking points

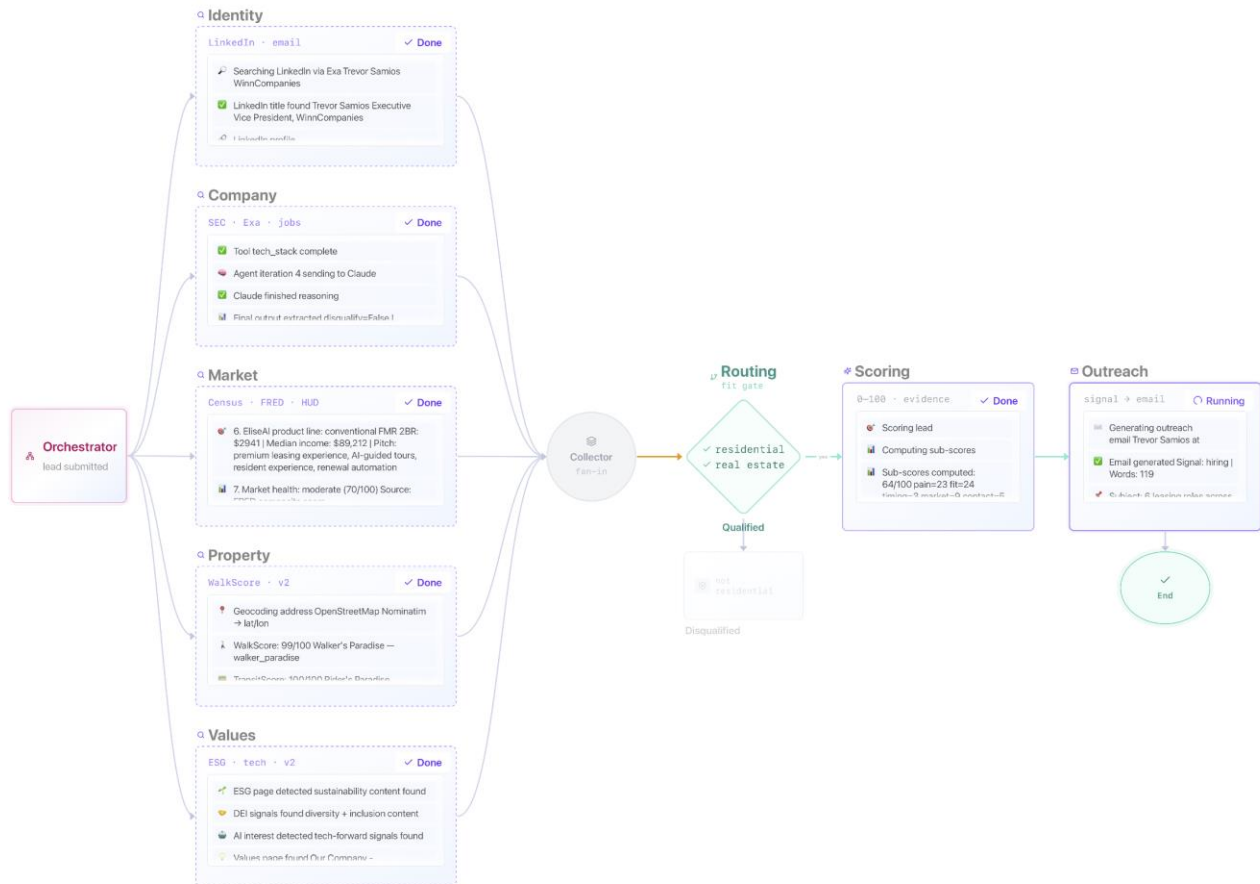
APIs Used and Why

API	Used For	Why This One
Exa AI	LinkedIn lookup, company about pages, ESG/values research	Best structured web search, returns clean text from specific pages. Better than Google for research tasks.
SEC EDGAR	Public company SIC codes, financials, 8-K filings	Free, authoritative, 100% accurate for public companies. SIC code is the most reliable ICP gate.

Adzuna Jobs API	Job posting data, leasing, maintenance, adoption roles	Real job data, city-specific search. Hiring pressure is the strongest and most verifiable pain signal.
Census ACS	City-level renter population, median income, vacancy	Government data, city-level accuracy. Determines market size and renter density.
FRED (Federal Reserve)	Unemployment, rent growth, population trends	Most authoritative economic source. Dynamic series search finds city-specific population data.
HUD Fair Market Rents	2BR FMR by metro area	Determines which EliseAI product to pitch: conventional, affordable, student, or single-family.
NewsAPI	Expansion signals, funding announcements	Free 30-day lookback. Catches major expansion and acquisition news.
WalkScore	Property walkability, transit, bike scores	Signals urban density and leasing velocity. High walkability = high prospect volume = need for 24/7 automation.

Section 3: Enrichment Pipeline

How The Pipeline Runs



Identity Agent

Searches LinkedIn via Exa AI using name + company. Extracts verified job title, infers seniority from title keywords (C-suite, VP, Director, Manager), and computes decision-maker probability from seniority + email pattern + corporate domain match. Falls back to email pattern analysis if LinkedIn not found, single initial email (j@company.com) signals founder, firstname.lastname signals mid-level professional.

Key limitation: common names (Sarah Johnson) sometimes match wrong LinkedIn profiles.

Production fix: LinkedIn Sales Navigator API with company + name combined search.

Company Agent (Agentic)

The most sophisticated agent. Uses Claude Sonnet with native Anthropic tool calling in a ReAct loop. Claude powered agent ,plans its own research rather than following hardcoded steps. Available tools: SEC EDGAR lookup, Exa private research, news search, job postings, tech stack detection.

Public companies (AvalonBay, Invitation Homes): SEC EDGAR returns structured financials, SIC code, and units managed in 1 iteration. Private companies (Greystar, Bozzuto): 3-4 iterations, EDGAR finds no filing, then Exa scrapes about page, Claude extracts units/revenue/AUM from raw text, then parallel tools gather news + jobs + tech stack. Non-residential companies (McDonald's SIC 5812): disqualified in 2 iterations — no wasted API calls on downstream research.

Critical extraction rules in system prompt: current units only (not historical milestones), revenue vs AUM distinction, is_residential gate. These rules prevent the most common LLM extraction errors.

Market Agent

Runs three services in parallel: Census ACS, FRED economic data, and HUD Fair Market Rents. Uses fuzzy city name scoring to match Census place names correctly, 'Philadelphia city, Pennsylvania' scores 100 vs 'New Philadelphia borough, Pennsylvania' scores 20, preventing the common census lookup bug.

Census ACS	Renter population, median income, vacancy rate, renter percentage. City-level data. Determines market size and renter density.
FRED	Unemployment rate (metro-level search), rent growth (national CUSR0000SEHA), population trend (dynamic series search finds DFWPOP, PAPHIL5POP etc).
HUD FMR	Fair Market Rent for 2BR by metro area. Determines EliseAI product line: FMR > \$1,800 = conventional, FMR < \$1,200 = affordable, university city = student housing, low renter density = single-family.
Leasing intensity	Composite score (0-100) combining unemployment, population growth, renter density, and renter pool size. High intensity = operators need 24/7 automated leasing NOW.

Property Agent

Geocodes the property address via OpenStreetMap Nominatim (free, no API key required), then fetches Walk Score, Transit Score, and Bike Score from WalkScore API. Classifies urban type from Walker's Paradise (90+) to Rural (<25). High walkability signals dense urban market with high prospect volume, strongest case for 24/7 leasing automation.

Values Agent

Runs three Exa AI searches in parallel: ESG and sustainability content, company values and mission page, and technology adoption signals. Detects AI interest, DEI commitment, and sustainability programs. Feeds into scoring as a soft signal for tech companies that are more receptive to automation tools.

Collector Node

A zero-cost gate node in LangGraph. Waits for all 5 parallel agents to complete, then checks the `is_residential` flag set by the company agent. Non-residential companies route to Disqualified immediately — scoring and outreach agents never run, saving LLM cost. Residential companies route to Scoring.

Scoring Agent

Hybrid approach: deterministic Python code computes five sub-scores from enriched data, then one Claude LLM call reviews the full context, adjusts for edge cases the formula misses, and generates the sales insights SDRs use in outreach.

Pain (0-30 pts)	Leasing + maintenance job openings, tech gap (no PropTech detected), portfolio size vs headcount mismatch
Fit (0-25 pts)	Units managed, public vs private, EliseAI-native integrations (Yardi, Entrata, OneSite, RealPage)
Timing (0-25 pts)	Systems adoption roles, SEC 8-K events, expansion news, funding signals, local hiring volume
Market (0-10 pts)	Leasing intensity score, market health composite, population trend
Contact (0-10 pts)	DM probability, seniority level, LinkedIn verified

The LLM call also generates: company snapshot (3 key facts), pain points, timing signals, pitch angle, integration hook, objection prep (2-3 Q&A pairs), and market context. These feed directly into the outreach email and the SDR's qualification brief.

Outreach Agent

One Claude LLM call. Selects the strongest available signal in priority order: hiring pressure → tech gap → growth signals → market pressure. Interprets the signal with operational insight specific to property management, not generic business observations. Generates a Signal → Insight → Soft Pitch → CTA email under 120 words.

The SDR can request a revision via the feedback loop in the UI. The agent accepts natural language feedback ('make it shorter', 'lead with market pressure', 'more casual tone') and regenerates applying the feedback exactly while maintaining the four-part structure.

Section 4: Lead Scoring Framework

Every lead is scored 0–100 across five weighted dimensions. The score reflects how urgently EliseAI should be in front of this operator , not just whether they're a fit, but whether now is the right moment.

Scoring uses a hybrid approach: deterministic Python code computes the five sub-scores from enriched data, then a single Claude LLM call reviews the full context, adjusts for edge cases the formula misses, and generates the sales insights SDRs use.

Score Tier Classification

Score	Tier	SDR Action
85–100	Hot	Contact within 24 hours. All signals aligned, priority outreach.
70–84	Warm	Prioritize outreach this week. Strong ICP, minor gaps.
55–69	Nurture	Add to the nurture sequence. Re-score in 30 days.
40–54	Low	Low effort only. Check back on expansion signals.
0–39	Pass / Cold	Disqualify or hold. No compelling signal today.

The Five Scoring Dimensions

1. Pain Score (max 30 pts) <i>How severe is the operational pain the prospect faces today?</i>	
Signals Evaluated	Scoring Bands
Active leasing hiring 2+ open leasing or maintenance roles indicates the team is stretched thin and responding manually to every inquiry.	Critical pain (25–30 pts) 3+ leasing roles open, no automation, large portfolio
Maintenance hiring pressure Multiple maintenance roles suggest high ticket volume with no automation routing requests to the right person.	Moderate pain (15–24 pts) 1–2 hiring signals or clear manual ops
No automation tooling If no PropTech is detected, the team is handling prospect communication entirely through email and phone.	Low pain (5–14 pts) Some tools in place, limited hiring
High portfolio size Large unit counts without proportional headcount means after-hours inquiries go unanswered, costing leases.	No signal (0–4 pts) Fully automated or no evidence of pain

2. Fit Score (max 25 pts)

How closely does the prospect match EliseAI's ideal customer profile?

Signals Evaluated	Scoring Bands
Residential focus EliseAI is purpose-built for residential property management. Mixed-use or commercial operators score lower.	Perfect ICP (20–25 pts) Residential, 200–2000 units, native integration detected
Unit count 50–5,000 units is the sweet spot. Too small = budget risk. Too large = enterprise complexity.	Strong fit (12–19 pts) Residential with partial integration match
Native integrations Yardi, Entrata, OneSite, or RealPage in the stack means a 30-day deployment with no rip-and-replace.	Partial fit (5–11 pts) Some ICP signals, gaps in size or stack
Company size 10–500 employee operators have budget authority and a clear decision-making structure.	Poor fit (0–4 pts) Commercial, micro-operator, or no stack signal

3. Timing Score (max 25 pts)

How ready is the prospect to evaluate and buy right now?

Signals Evaluated	Scoring Bands
Active expansion Acquiring properties or entering new markets means new leasing capacity is needed immediately.	Buy now (20–25 pts) Active expansion + tech gap + recent growth event
Tech gap with existing PropTech Using Yardi or Entrata but missing leasing automation means they already believe in PropTech — just haven't found EliseAI yet.	Near-term (12–19 pts) 1–2 strong timing signals
Recent funding or growth event Fundraises and major portfolio announcements signal budget availability and a window for new tooling decisions.	Possible (5–11 pts) Stable operator, moderate signals
Lease-up phase New construction in lease-up needs to fill units fast — automation directly drives revenue velocity.	Not ready (0–4 pts) No timing indicators detected

4. Market Score (max 10 pts)

How much does the local leasing market intensify the need for automation?

Signals Evaluated	Scoring Bands
High leasing intensity city	High intensity (8–10 pts)

Markets like Dallas, Austin, Phoenix, and Charlotte have prospects shopping 3–4 properties simultaneously. Response speed wins the lease.	Top-tier market, high leasing velocity
Competitive vacancy Sub-5% vacancy markets mean operators are managing high inbound volume without extra staff.	Moderate (5–7 pts) Growing or competitive secondary market
Seasonal surge Spring/summer leasing seasons dramatically increase inquiry volume for unprepared teams.	Low intensity (0–4 pts) Stable or low-velocity market

5. Contact Score (max 10 pts) <i>How decision-ready is the specific contact identified?</i>	
Signals Evaluated	Scoring Bands
Title match VP of Operations, Director of Leasing, Regional Manager, or COO titles have direct budget authority for operational tooling.	Decision maker (8–10 pts) C-suite or Director+, verified email, active LinkedIn
Verified email A confirmed, deliverable email address avoids bounces and signals the contact is active.	Influencer (5–7 pts) Manager-level with verified contact
LinkedIn presence An active LinkedIn profile confirms the contact is real, currently employed, and reachable for follow-up.	Weak contact (0–4 pts) Generic email, unverified title, or low seniority
Decision-maker level C-suite and Director+ contacts can approve deals. Manager-level contacts can champion but may need escalation.	

Scoring Architecture Note

The five sub-scores are computed deterministically in Python , no LLM is needed for the formula. A single Claude Sonnet call then reviews all enrichment data, adjusts for edge cases the formula misses, and generates the sales insights (pitch angle, objection prep, talking points) SDRs use for outreach. This hybrid approach gives consistent, explainable scores with LLM quality narrative intelligence on top.

Section 5: Outreach Generation

After scoring, the Outreach Agent generates a personalized cold email using a four-part structure proven to drive replies in B2B SaaS sales. The SDR receives a ready-to-send email — not a template to fill in.

Email Structure: Signal → Insight → Soft Pitch → CTA

Step	Purpose	Example (Greystar, Scott Berka)
1. Signal	Open with ONE specific, verifiable signal from enrichment data	"Saw Greystar has 4 leasing roles and 7 maintenance techs open right now"
2. Insight	Interpret what that signal means for them operationally — show domain expertise, not generic observations	"At 1.1M units, that hiring volume means your best leasing agents are spending 80% of their day answering the same 10 questions instead of converting tours"
3. Soft Pitch	Connect EliseAI naturally — don't hard sell, reference proof points	"EliseAI automates that inquiry triage. AvalonBay runs this across 290+ properties alongside Yardi — deploys in 30 days, no rip-and-replace"
4. CTA	One low-friction ask — not a demo request, a conversation	"Worth 15 min to see what this looks like operationally for a portfolio at your scale?"

Signal Priority

The agent selects the strongest available signal to lead with, in this order:

- ☐ Hiring pressure : strongest signal, most specific, hardest to fake interest in
- ☐ Tech gap : using PropTech but not EliseAI specifically
- ☐ Growth signals : expansion, funding, new markets
- ☐ Market pressure : high leasing intensity in their city

SDR Feedback Loop

If the SDR isn't satisfied with the generated email, they can request a revision directly in the UI. The system accepts natural language feedback:

- ☐ "Make it shorter , under 80 words"
- ☐ "Lead with market pressure instead of hiring"
- ☐ "More casual tone"

The Outreach Agent regenerates the email applying the feedback exactly, maintaining the Signal → Insight → Pitch → CTA structure. This loop can repeat until the SDR is satisfied — no regeneration limit.

Why This Structure Works

Signal: Shows you did real research , not a spray-and-pray template

Insight: Demonstrates operational understanding of property management ,builds credibility fast

Pitch: References real customers (AvalonBay, Equity Residential) + specific integration story

CTA: Low friction , '15 min' not 'book a demo' ,dramatically improves reply rates

Section 6: Automation

LeadAI supports two automation modes: a daily scheduled queue kick and an on-demand trigger via form submission or CSV upload. Both feed the same DB-backed pipeline queue.

Trigger: On-Demand (Form + CSV Upload)

The primary trigger for the demo ,SDRs submit leads instantly.

- ☐ Single lead: submitted via the LeadAI dashboard form
- ☐ Bulk leads: uploaded as CSV via the bulk import button
- ☐ Both create leads with status = queued in PostgreSQL immediately
- ☐ POST /api/leads returns instantly — pipeline runs in background
- ☐ Frontend connects to SSE stream for live agent updates

Schedule: Daily Queue Kick (APScheduler)

A background scheduler automatically processes queued leads on a configurable schedule.

- ☐ Default: daily at 9:00 AM local time
- ☐ Configurable: SDR managers can update hour/minute from the Settings page in the dashboard
- ☐ Uses APScheduler AsyncIOScheduler ,runs inside the FastAPI process
- ☐ On schedule trigger: worker claims oldest queued lead and begins pipeline

Settings Page

The LeadAI dashboard includes a Settings page where the scheduler timing can be updated.

Sales managers can change the daily run time without touching code.

Toggle: pause/resume the scheduler entirely for maintenance windows.

Queue System

The pipeline queue ensures leads are processed reliably without race conditions.

Feature	Implementation
One active pipeline	Single asyncio worker — one lead runs at a time, others wait in queue. Prevents API rate limit issues.
DB row locking	PostgreSQL SELECT ... FOR UPDATE SKIP LOCKED — safe queue claiming even if multiple processes run.
Retry logic	Failed pipelines retry up to 2 times before marking as failed. Prevents data loss on transient API errors.
Stalled recovery	Processing leads older than 30 minutes are automatically requeued on nextstartup. Handles server restarts gracefully.
SSE event replay	All pipeline events are persisted to DB. If the browser refreshes mid-pipeline, the SSE stream replays from the last event.
Status lifecycle	queued → processing → complete / disqualified / failed. Each state visible in the dashboard lead list.

Lead Status Lifecycle

queued	Lead submitted, waiting for worker to pick up
processing	Pipeline actively running — agents enriching in parallel
complete	All 7 agents finished — score, email, and insights ready
disqualified	Company agent determined not residential — no further processing
failed	Pipeline error after 2 retry attempts — SDR notified

Section 7: Test Cases & Validation

LeadAI was tested against 16 real leads spanning enterprise public REITs, large private operators, student housing, affordable housing, single-family rental, and intentional disqualifications. Below are the some examples, showing the working of LeadAI.

A full results CSV is also attached. Below are some of the test cases that had some interesting finds.

Case 1: Greystar — HOT Lead (85/100)

Demonstrates the private company agentic research path and enterprise scoring.

Scott Berka Greystar · Sr. Managing Director, Brand & CX · Dallas, TX	
HOT 85/100 · Contact within 24 hours	
Company path	Private company → SEC EDGAR returned no filing → Exa scraped greystar.com/about → Claude extracted 1.1M units, \$600M revenue, \$320B AUM in 3 agentic iterations
Score breakdown	Pain: 16 (4 leasing + 7 maintenance roles open) · Fit: 24 (1.1M units, enterprise, Yardi+Entrata+OneSite) · Timing: 8 · Market: 9 (Dallas high intensity) · Contact: 9 (DM prob 1.0, VP-level)
Identity result	LinkedIn confirmed: Sr. Managing Director Brand & Customer Experience. DM probability: 1.0. LinkedIn: linkedin.com/in/scottberka
Tech stack	Yardi, Entrata, OneSite — all three EliseAI-native integrations detected via job posting keyword analysis
Recommended action	Immediate executive outreach within 24 hours. Lead with Greystar scale (1.1M units) + CX focus = EliseAI AI-guided tours. Request 15-min intro call to discuss resident experience automation across Yardi/Entrata/OneSite portfolio.
Signal used	Hiring pressure — 4 leasing + 7 maintenance roles open in Dallas. Email subject: '11 open roles across leasing + maintenance teams'

Case 2: McDonald's — Disqualified (SIC 5812)

Demonstrates intelligent early disqualification — scoring and outreach agents never ran.

John Wendel / Chris Kempczinski McDonald's Corporation · Chicago, IL	
DISQUALIFIED 6/7 agents ran · Scoring + Outreach never triggered	
Disqualification reason	SEC EDGAR returned SIC 5812 (Retail-Eating Places). Company agent stopped in 2 iterations — no wasted API calls on news, jobs, or tech stack research.

System message	Do not contact — Not a residential real estate company. McDonald's is a restaurant chain (SIC 5812: Retail-Eating Places). EliseAI only serves multifamily apartments, student housing, affordable housing, and single-family rental operators.
Pipeline efficiency	Disqualified in ~15 seconds. Scoring and outreach agents never ran — zero LLM cost on non-ICP lead. Only 6 of 7 agents completed.
Why this matters	Without disqualification logic, SDRs would waste 90 minutes manually researching McDonald's before realizing it's not a fit. LeadAI eliminates this at API cost of <\$0.01.

Case 3: WinnCompanies — HOT (87/100)

Demonstrates affordable housing product line detection and correct ICP classification.

Trevor Samios WinnCompanies · EVP Operations Strategy & Solutions · Boston, MA	
HOT 87/100 · Contact within 24 hours	
Product line detected	Affordable housing — HUD FMR + Census income data identified Boston as affordable/workforce market. EliseAI pitch: multilingual support (51 languages), fair housing compliance automation.
Why HOT at 87	EVP-level contact + 97K+ unit portfolio + Yardi/RealPage integrations + Boston high leasing intensity + 6 open leasing roles = strong pain + fit + contact quality.
Recommended action	Contact Trevor Samios (EVP) within 24 hours. Lead with affordable housing compliance value prop + staffing relief. Reference Yardi/RealPage native integrations for zero-friction implementation.
Interesting signal	System correctly identified WinnCompanies as residential despite affordable housing focus — demonstrates ICP precision beyond just 'luxury' multifamily.

Case 4: Real McCoy Group — Disqualified (Edge Case)

Demonstrates ICP precision — distinguishes property managers from real estate agents.

Ryan McCoy Real McCoy Group (Coldwell Banker) · Bethany Beach, DE	
DISQUALIFIED Real estate agent, not property manager	
Disqualification reason	Real estate brokerage/sales team, not a residential property management company. EliseAI serves property managers who operate multifamily/apartment communities, not real estate agents who buy/sell homes.
Why this is important	This is a subtle but critical distinction. Both involve 'real estate' but EliseAI's ICP is operators who manage recurring resident relationships — not transactional home sales agents.

System accuracy	Company agent correctly identified the company as a brokerage despite the address being residential (beach community). SIC code and Exa about page analysis caught this.
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Case 5: Equinix & Simon Property — Disqualified (Non-Residential Real Estate)

Demonstrates that the system understands EliseAI's ICP is specifically residential — not all real estate.

Nicole Collins (Equinix) + Natalie Turpan (Simon Property Group) Data Center REIT + Retail Mall REIT — both disqualified correctly	
DISQUALIFIED Both are REITs but not residential	
Equinix	Data center REIT (SIC 6798 — Real Estate Investment Trusts, but infrastructure). is_residential=False triggered by SEC EDGAR SIC analysis. Disqualified in 2 iterations.
Simon Property	Retail mall REIT. SIC code identified as retail/commercial real estate. is_residential=False. Disqualified before reaching scoring or outreach.
ICP precision	Both are legitimate real estate companies — a naive system might contact them. LeadAI understands EliseAI's ICP is specifically residential operators: multifamily, student, affordable, single-family rental.

Section 8: Rollout Process

Rolling out LeadAI follows a three-phase approach: piloting with a small SDR cohort, validate and iterate, then scale across the full sales org. Each phase has clear entry criteria and success metrics before proceeding.

Phase 1: Shadow Mode Pilot (Weeks 1)

Before changing any SDR workflow, LeadAI runs in parallel with the existing process. SDRs conduct their normal research while LeadAI enriches the same leads simultaneously. At the end of each day, results are compared side-by-side.

Why shadow mode first:

- ☐ Zero workflow disruption, SDRs continue their current process
- ☐ Creates a ground truth dataset for accuracy validation
- ☐ Builds SDR trust before asking them to rely on the tool
- ☐ Identifies gaps in enrichment quality before full commitment

Phase 1 Entry Criteria

- ✓ LeadAI deployed on staging environment
- ✓ 2–3 SDR volunteers identified for pilot cohort
- ✓ Baseline SDR research time measured (avg. minutes per lead)
- ✓ Success metrics agreed upon with Sales Leadership

Phase 2: Active Pilot [2–3 SDR Cohort (Weeks 2–3)]

SDRs in the pilot cohort begin using LeadAI as their primary research tool. They receive enriched leads with scores, insights, and draft emails, and provide structured feedback after each outreach.

What the SDR workflow looks like:

- ☐ New lead submitted via LeadAI form or CSV bulk upload
- ☐ Pipeline runs in 90 seconds and SDR is notified when one batch is complete.
- ☐ SDR reviews score, insights, and draft email
- ☐ SDR edits email if needed using the feedback loop (regenerate button)
- ☐ SDR sends outreach via Outreach.io with one click
- ☐ SDR flags any enrichment errors for the GTM Engineer to review

Phase 2 Success Metrics

- ✓ SDR research time reduced by ≥50% per lead

- ✓ Lead score accuracy validated against SDR judgment: ≥80% agreement
- ✓ Outreach email quality rated ≥7/10 by SDRs on average
- ✓ Disqualification accuracy: ≥95% (non-residential companies correctly filtered)
- ✓ Zero false disqualifications of residential operators. Important to loose out real good leads.

Phase 3: Full Sales Org Rollout (Weeks 4–6)

After the pilot validates accuracy and SDR satisfaction, LeadAI rolls out to the full sales team. This phase includes Salesforce integration so enriched leads auto-populate as accounts, and Outreach.io integration for automated sequence enrollment.

Rollout steps:

- ❑ 30-minute training session for all SDRs (Very simple to use project) with live demo + Q&A.
- ❑ Written playbook: How do we identify wrong edge cases popping up?
- ❑ Salesforce custom fields mapped to LeadAI enrichment outputs
- ❑ Outreach.io sequence auto-enrollment based on tier (If hot, then immediate, if warm, then can wait for 72 hours.)
- ❑ Monthly performance review with Sales Leadership

Section 9: Timelines

The rollout follows an 8-week plan from first pilot lead to full org adoption. Each phase gates on measurable success criteria before proceeding.

Timeline	Phase	Key Activities
Week 1	Shadow Mode Setup and Shadow Mode Validation	Deploy to staging, onboard 2–3 SDR volunteers, run parallel enrichment on live leads, measure baseline research time. Compare LeadAI output vs SDR manual research, identify accuracy gaps, fix enrichment issues, validate scoring framework
Week 2	Active Pilot Begins	Pilot SDRs switch to LeadAI as primary tool, collect structured feedback daily, track research time reduction

Week 3	Pilot Iteration	Fix top 3 SDR pain points, retune scoring weights based on SDR feedback, validate disqualification accuracy
Week 4	Salesforce Integration	Map LeadAI enrichment fields to Salesforce custom fields, test bidirectional sync, set up Outreach.io sequence triggers
Week 5	Full Team Training	30-min training session, written playbook distributed, all SDRs onboarded to LeadAI
Week 6	Full Org Rollout	All SDRs using LeadAI, GTM Engineer monitoring adoption, weekly check-ins with Sales Leadership
Ongoing	Iteration & Optimization	Monthly scoring weight reviews, A/B test outreach email structures, expand to new lead sources

Go/No-Go Criteria Before Each Phase

Shadow → Active Pilot: Enrichment accuracy ≥80%, zero critical bugs in disqualification logic

Active Pilot → Full Org: SDR research time -50%, email quality ≥7/10, SDR NPS ≥7

Full Org → Ongoing: Salesforce sync working, Outreach.io sequences triggering correctly

Section 10: Stakeholders

LeadAI touches multiple teams across EliseAI's GTM organization. Each stakeholder has a distinct role in the rollout, from technical validation to adoption coaching.

Stakeholder	Role	How They're Involved
CEO / Leadership	Executive Sponsor	Approves rollout strategy, sets adoption expectations, reviews monthly ROI report
Sales Leadership	Business Owner	Defines success metrics, approves scoring framework assumptions, champions adoption with SDR team
SDR Team	Primary Users	Daily tool users, provide feedback on enrichment quality and email drafts, flag errors
SDR Manager	Change Manager	Communicates rollout to team, enforces adoption, monitors SDR feedback
Revenue Operations	Systems Owner	Manages Salesforce field mapping, Outreach.io integration, data governance
GTM Engineer	Builder & Owner	Builds, maintains, and iterates on LeadAI, primary point of contact for all technical issues

Marketing	Lead Source Partner	Provides inbound lead list from campaigns, validates enrichment matches marketing intent data
IT / Security	Compliance	Reviews API key management, data privacy for third-party APIs, access controls

Communication Plan

Each stakeholder group receives tailored communication throughout the rollout:

- ❑ Sales Leadership: Weekly 30-min sync during pilot, monthly thereafter. Topics: accuracy metrics, SDR adoption rate, ROI tracking.
- ❑ SDR Team: Daily Slack channel for feedback during pilot. Written playbook on day 1 of full rollout.
- ❑ Revenue Operations: Technical specifics document for Salesforce field mapping and Outreach.io sequence logic before Week 5.
- ❑ CEO / Leadership: Two page monthly ROI summary: leads enriched, time saved, reply rate improvement, disqualification accuracy.

Section 11: Production Upgrades

LeadAI was built in 5 days using easily available free tier APIs and a local PostgreSQL database. The architecture here is deliberately designed to scale ,each component has a clear production upgrade path that doesn't require rebuilding the system from scratch.

API & Data Quality Upgrades

Component	Current (Demo)	Production Upgrade	Impact
Identity Lookup	Exa AI LinkedIn search (~80% hit rate on common names)	LinkedIn Sales Navigator API ,verified title, company, seniority with 95%+ accuracy	Eliminates wrong-person matches on common names. Enables leadership change alerts.
Company Research	Exa about page scraping + Claude extraction	Clearbit / ZoomInfo API,structured firmographic data with verified employee count, revenue, tech stack	Improves on LLM extraction. Adds intent signals not available via scraping.

Hiring Signals	Adzuna Jobs API (free, 20 results max per search)	Burning Glass / Lightcast API, full job market data, historical hiring trends, skills demand	10x more job data. Enables trend analysis: is hiring accelerating or decelerating?
Market Rent Data	FRED national rent index (not city-specific)	CoStar API , city-level rent growth, vacancy rates, absorption rates by submarket	City-level rent pressure directly improves leasing intensity scoring accuracy.
Tech Stack	Job posting keyword matching (indirect signal)	BuiltWith API (\$295/mo),verified installed technologies per company domain	Confirms Yardi/Entrata/OneSite usage directly. Eliminates false positives from job post scanning. Significantly better lead if they use these technologies.
Buying Intent	NewsAPI free tier (30-day lookback, limited sources)	Bombora Intent Data,tracks research behavior across 5,000+ B2B sites	Identifies companies actively researching property management software before they contact sales.

Infrastructure Upgrades

Queue System

Current: Single asyncio worker with DB row locking , processes one lead at a time. Sufficient for pilot volume (10–50 leads/day).

Production: Celery + Redis distributed queue with multiple workers. Enables horizontal scaling here, 10 workers can process 10 leads simultaneously, reducing per lead latency from 90 seconds to near real time at scale.

Database

Current: Local PostgreSQL (Docker). I am handling the schema migrations handled manually via startup scripts.

Production: Managed PostgreSQL on AWS RDS or Railway. With formal Alembic migrations, every schema change is versioned, reviewable, and rollback-safe. Read replicas for the analytics dashboard.

Authentication & Security

Current: No authentication, local development only.

Production: API key authentication for all endpoints. Role-based access: SDRs can submit leads and view results; Sales Managers can view analytics; GTM Engineer has admin access. Rate limiting per API key to prevent abuse.

CRM & Sales Tool Integrations

The highest impact production upgrade is direct integration with EliseAI's existing sales stack:

- ❑ Salesforce: When a lead completes enrichment, LeadAI auto creates or updates the Salesforce Account and Contact record. All enrichment data maps to custom fields: LeadAI_Score__c, LeadAI_Tier__c, LeadAI_PitchAngle__c, LeadAI_EmailDraft__c. Hot leads trigger a Salesforce task assigned to the owning SDR.
- ❑ Outreach.io: Pipeline completion triggers automatic sequence enrollment based on tier. For hot leads there is high-touch immediate sequence. With warm leads, a 72-hour nurture sequence. Disqualified leads are suppressed from all sequences automatically.
- ❑ Slack notifications: When a hot lead (80+) completes enrichment, LeadAI posts a rich notification to #hot-leads Slack channel with score, key facts, and one-click link to the draft email.

Extending to Healthcare Vertical

EliseAI serves both housing and healthcare. I have currently built the tool to handle housing leads as per the assignment scope. The LeadAI pipeline can be built to extend, the same 7-agent architecture applies to healthcare leads with different ICP signals:

- ❑ Identity agent: same (LinkedIn + email pattern)
- ❑ Company agent: healthcare SIC codes, practice size, EHR systems (Epic, Cerner, athenahealth) instead of PropTech
- ❑ Market agent: CMS data (Medicare/Medicaid reimbursement rates) instead of Census/HUD
- ❑ Scoring agent: different dimension weights, timing signals around value-based care transitions, EHR adoption rates, practice acquisition activity
- ❑ Outreach agent: same Signal -> Insight -> CTA structure, different domain knowledge

This extension means LeadAI becomes EliseAI's company-wide lead intelligence platform -> not a one-off SDR tool.

Automated Re-Scoring for Nurture & Low Leads:

Currently, Nurture (55–69) and Low (40–54) leads are scored once and manually re-evaluated. In production, LeadAI would automatically re-enrich and re-score these leads on a schedule, turning the pipeline into a living lead database, not a one-time snapshot.

Tier	Current	Production Behavior	Why
Nurture (55–69)	Scored once. SDR manually decides when to re-check.	Automatically re-enriched every 30 days. If new expansion signals, hiring surge, or funding detected, re-scored and promoted to Warm or Hot.	Nurture leads often become buyers after a portfolio expansion or leadership change. Catching this automatically means SDRs never miss the window.
Low (40–54)	Scored once. Sits in the database indefinitely.	Re-enriched every 60 days. Only promoted if score crosses 55. Otherwise stays Low with updated signal log.	Low leads are usually too small or wrong timing, not wrong company. Periodic re-scoring catches when they grow into ICP or enter an expansion phase.

How Re-Scoring Works in Production:

1. APScheduler runs nightly, checks all Nurture leads last enriched > 30 days ago
2. Re-runs company agent (new job postings, expansion news, SEC filings)
3. Re-runs market agent (updated Census/FRED data)
4. Recomputes score with fresh data
5. If score crosses tier boundary -> SDR notified via Slack #lead-updates
6. If score unchanged -> lead updated silently, re-check scheduled

This turns LeadAI from a one-shot enrichment tool into a living lead intelligence system. SDRs stop manually tracking "check back in 30 days" notes, the system does it for them.

ROI Tracking Framework:

In production, LeadAI tracks these metrics weekly to demonstrate value and guide iteration:

- ☐ SDR research time per lead: baseline vs LeadAI (target: -50%)
- ☐ Outreach reply rate by tier: Hot vs Warm vs Nurture (validates scoring accuracy)
- ☐ Time from lead submission to first outreach (target: <10 minutes vs current ~24 hours)
- ☐ Disqualification accuracy: % of disqualified leads that were actually non-residential
- ☐ Email edit rate: % of AI generated emails sent with no edits (higher = better quality)
- ☐ Pipeline influenced: revenue from leads first touched by LeadAI

Expected ROI at Full Scale (150 Sales Employees, around 30 SDRs)

Research time saved: 30 SDRs × 45 min/lead × 10 leads/day = 225 hours/day

At \$50/hr fully loaded: \$11,250/day in recovered SDR capacity

Faster response time: <10 min vs 24 hrs = significantly higher prospect conversion

Disqualification value: SDRs stop wasting time on non residential or invalid leads.
API costs at scale: \$0.08-0.12 per lead enrichment (with batch processing.)

LeadAI — Built for EliseAI GTM Engineer Assignment